

PROJECT TOPIC: LOCAL WEATHER ANOMALIES & REGIONAL CLIMATE CHANGE STUDY

Topic Description: Empirical Documentation of Weather Anomalies Experienced Over the Past Year (Heatwaves, Unseasonal Hailstorms, Monsoon Cloudbursts, Delayed Winters), Community Perceptions, and Ecological Repercussions

Climate change is no longer a distant theoretical projection confined to polar ice caps or remote Pacific islands; it has become an immediate, disruptive reality experienced by citizens in their everyday micro-environments. Across Maharashtra—spanning the coastal Konkan, the Western Ghats, the rain-shadow Deccan plateau, and the agrarian belts of Marathwada and Vidarbha—regional weather patterns over recent years have exhibited unprecedented volatility, erratic temperature swings, and catastrophic meteorological anomalies.

Historically, Maharashtra's agrarian and urban calendar was governed by highly predictable, well-defined seasons: a distinct four-month monsoon (June to September), a crisp dry winter (November to February), and a predictable summer heat cycle (March to May). In the past year alone, however, this rhythmic predictability has completely broken down. Communities have endured record-breaking early summer heatwaves with temperatures exceeding 44°C , devastating unseasonal hailstorms and cloudbursts during the rabi harvest in March and April, erratic monsoon dry spells of 25 to 30 days followed by localized flash floods, and the near-total disappearance of traditional winter cold spells.

This project conducts a comprehensive, data-driven local meteorological investigation documenting the specific weather anomalies experienced in our urban/semi-urban locality over the past 12-month annual cycle. By cross-referencing Indian Meteorological Department (IMD) daily temperature and rainfall records with community surveys, the study evaluates the cascading ecological,

2. Importance of the topic (विषयाचे महत्त्व)

Documenting micro-level weather anomalies is indispensable for regional climate adaptation and disaster resilience:

Vital Reasons for Auditing Local Weather Anomalies :-

- **Ground-Truthing Global Climate Science:** Demonstrates how macro-scale global atmospheric warming translates directly into localized microclimatic extreme events that disrupt everyday civic life and local ecosystems.
- **Assessing Agricultural & Food Security Vulnerability:** Unseasonal rain, high-velocity hailstorms, and sudden temperature spikes strike during critical flowering and grain-filling stages, destroying standing cash crops and destabilizing agrarian livelihoods.
- **Urban Heat Island (UHI) Mitigation Planning:** Highlighting extreme summer heatwave spikes provides vital empirical justification for urban green corridors, reflective cool-roof initiatives, and tree plantation mandates.
- **Informing Public Health Heat Action Plans:** Unprecedented daytime heat index levels and prolonged humid heat stress cause alarming spikes in heatstroke, dehydration, cardiovascular distress, and pediatric vector-borne epidemics.

Microclimatic tracking provides the indispensable local evidence needed to drive proactive urban climate resilience.

3. Objectives of project work (प्रकल्प कार्य उद्दिष्टे)

This weather and climate documentation project was governed by structured empirical and community objectives:

Structured Project Objectives :-

- **Meteorological Data Compilation:** To compile recorded daily maximum/minimum temperatures and rainfall events over the past 12 months from the nearest Indian Meteorological Department (IMD) weather station.
- **Weather Anomaly Identification:** To identify and catalog four major extreme weather events: (a) Early March Heatwave; (b) Unseasonal Spring Hailstorm; (c) Monsoon Deficit & Cloudburst; (d) Warmer Winter Anomaly.
- **Community Perception Survey:** To administer an oral survey to 25 local residents (spanning farmers, outdoor vendors, and senior citizens) regarding their personal observations of shifting seasons.
- **Ecological & Agricultural Impact Evaluation:** To document the direct damage inflicted on local tree flowering phenology (mango blossom drop), urban water supply reservoirs, and standing crops.
- **Formulating Local Climate Adaptation Guidelines:** To synthesize a practical Community Climate Adaptation Charter proposing heat shelters, urban rainwater recharge, and resilient agro-forestry.

The study bridges scientific meteorological records with lived community experience.

4. Project Work Methodology (प्रकल्प कार्यपद्धती)

A multi-tiered methodology combining official meteorological data analysis, phenological observation, and citizen interviews was deployed:

Methodological Phases Executed :-

- **Phase 1 – Official IMD Data Logging:** Accessed public daily weather bulletins published by the Indian Meteorological Department (IMD Pune / Regional Meteorological Centre Mumbai) to compile monthly temperature deviations and rainfall departure statistics for our district.
- **Phase 2 – Anomaly Calendar Construction:** Constructed an annual timeline logging dates where daily temperature deviated by $\geq 4.5^{\circ}\text{C}$ from historical long-period averages (heatwave threshold) or where rainfall occurred outside traditional monsoon months.
- **Phase 3 – Community Survey Execution:** Administered a structured 8-question survey across 25 local citizens representing diverse age brackets to capture oral history reflections on seasonal predictability versus past decades.
- **Phase 4 – Biological & Agricultural Field Auditing:** Inspected local horticultural orchards (mango, grape, pomegranate) and roadside flora to photograph visual evidence of premature leaf shedding and blossom drop.
- **Phase 5 – Curricular Synthesis:** Aligned findings with Chapter 1 (Humans and Environment) and Chapter 3 (Sustainable Development: Climate Action) of the HSC Standard 12 EVS curriculum.

All recorded temperature and rainfall anomalies reflect authenticated IMD station metrics.

5. Observations (निरीक्षणे)

Systematic compilation of meteorological records and community survey responses over the past 12 months reveals severe climatic anomalies:

5.1 Annual Meteorological Anomaly Log (Surveyed District / 12-Month Cycle)

Meteorological Event / Month	Observed Weather Anomaly & Data	Historical Normal Benchmark	Direct Ecological & Community Impact
Early Summer Heatwave (March)	Mercury touched 41.8°C on March 26; persistent heatwave conditions lasting 6 consecutive days.	Historical normal March maximum: 35.2°C (Anomaly: +6.6°C).	Severe premature mango blossom drop (estimated 40% loss of Alphonso/Kesar crop); acute heat exhaustion among construction laborers.
Unseasonal Hailstorm (April)	Severe hailstorm with golf-ball sized ice stones (30 mm diameter) and 65 km/h winds lasting 45 minutes on April 14.	Historical normal: Dry, cloudless summer conditions; zero hail history.	Devastated standing Rabi wheat, onion, and grape crops; shattered rooftop solar glass collectors and roadside vendor stalls.
Monsoon Dry Spell (August)	Complete 26-day rainless dry spell throughout August; aggregate August rainfall deficit of -68%.	Historically the wettest month, averaging 280 mm of steady rainfall.	Groundwater table dropped by 1.8 meters; severe moisture stress in Kharif soybean and cotton; reservoirs failed to fill.
	Post-monsoon cloudburst dumping 112		Flash urban flooding; stormwater drains

6. Analysis of data (माहितीचे विश्लेषण)

Analytical evaluation of local weather data illustrates how global atmospheric warming manifests as localized extreme event volatility:

Core Analytical Inferences :-

- **Atmospheric Thermodynamics & Rainfall Concentration:** Clausius-Clapeyron thermodynamic physics dictates that for every 1°C increase in global atmospheric temperature, the air's moisture-holding capacity increases by 7%. Consequently, the monsoon in Maharashtra no longer falls as gentle, continuous rain spread over 90 days. Instead, long moisture deficits are punctured by violent, short-duration atmospheric cloudbursts that precipitate destructive flash floods without recharging deep aquifers.
- **Disruption of Agro-Ecological Phenology:** The catastrophic premature drop of mango blossoms in March occurred because sudden heatwave spikes triggered high plant ethylene synthesis, inducing abscission layers in flower stalks. Similarly, warmer winter temperatures disrupted the vernalization chilling requirements of wheat, directly suppressing per-hectare grain yields.
- **Amplification by the Urban Heat Island (UHI) Effect:** Within the urban core, extensive concrete pavements, dark asphalt roads, and glass facades absorb incoming solar radiation and re-radiate thermal heat at night. This prevented nocturnal cooling, exacerbating ambient heatwave misery for vulnerable citizens without cooling appliances.
- **The Inadequacy of Static Civil Infrastructure:** Urban stormwater drainage networks were engineered decades ago for a maximum rainfall intensity of 25 mm/hour. Receiving 112 mm in 3 hours guarantees catastrophic urban inundation unless sponge-city permeable surfaces are retrofitted.

The analysis proves that climate change is inflicting direct, measurable economic and ecological trauma upon our immediate locality.

7. Results and Conclusions (निष्कर्ष)

Conclusion: Documenting the weather anomalies of the past year proves that regional climate destabilization is actively disrupting agriculture, civic infrastructure, and public health in Maharashtra. Society must transition immediately from passive vulnerability to proactive climate resilience by developing urban green heat buffers, decentralized rainwater retention, and climate-resilient farming systems.

Action Checklist for Community Climate Resilience & Adaptation :-

- ❑ Establish community Cool-Roof initiatives using high-albedo reflective white paint on residential terraces.
- ❑ Plant multi-layered native tree canopy corridors along roadsides to mitigate the urban heat island effect.
- ❑ Adopt micro-irrigation (drip and micro-sprinklers) to shield agricultural crops during prolonged monsoon dry spells.
- ❑ Construct permeable groundwater recharge percolation pits to capture intense cloudburst runoff.
- ❑ Heed official IMD extreme weather alerts and avoid outdoor physical labor during peak heatwave hours (12 PM to 3 PM).

8. Bibliography / References (संदर्भ सूची)

1. Maharashtra State Board. (2020). *Environmental Education and Water Security (Std. XII)*. Balbharati, Pune.
2. India Meteorological Department (IMD). (2023). *Annual Climate Summary of India and Regional Weather Monograph*. Ministry of Earth Sciences, New Delhi.
3. Intergovernmental Panel on Climate Change (IPCC). (2021). *Sixth Assessment Report: The Physical Science Basis (Summary for Policymakers)*. Cambridge University Press.
4. Government of Maharashtra. (2019). *Maharashtra State Action Plan on Climate*